

NumPy's public API, rebuilt on safe Rust storage

A different Python import is acceptable. The destination is the same public functionality and defined behavior, validated release by release.

0.1 IMPLEMENTED LOCALLY

HOSTED RELEASE GATE PENDING

48

preview cases passed locally
CPython 3.12-3.14 + NumPy 2.5.3

7 / 7

storage tests pass under Miri
macOS ARM64

12

wheel targets in release gate
4 platform targets x 3 Python versions

0.1 PREVIEW SCOPE

- **Values:** bool, int64, uint64, float32, float64; preserve large integers, NaN, signed zero, and float32 warnings.
- **Views:** shape/dtype/size/byte strides; integer and slice indexing, negative steps, and nested shared views.
- **Writes:** scalar assignment, exact-shape same-dtype Raptors-array assignment, overlap snapshots, independent copies.
- **Runtime:** initialized Rust-owned buffers; NumPy is a locked test oracle, not a runtime dependency.

RELEASE GATE

- Build and test CPython 3.12-3.14 wheels for Linux x86-64, macOS Intel/ARM64, and Windows x86-64.
- Pass hosted Miri and Linux AddressSanitizer checks; install each wheel and test with the pinned NumPy oracle.
- Run the manual build-only workflow on the candidate. Publish only from a passing exact vX.Y.Z tag through PyPI trusted publishing.

0.x RELEASE PHASES

0.1-0.3	safe preview, dtype/layout foundation, numeric ufuncs
0.4-0.6	array routines, scientific algorithms, files and interoperability
0.7-0.9	specialized data models, public API completion, conformance closure

FIRST PYTHON-LEVEL BASELINE / NO SPEED CLAIM

On 250,000 int64-compatible values, NumPy had lower median latency for all four measured calls: create 3.75 ms vs 7.68 ms; slice 0.67 us vs 0.96 us; overlapping assignment 34.2 us vs 7.56 ms; copy 112 us vs 4.25 ms. One-host results are informational; native RSS is coarse.

Full API compatibility and a general memory-safety guarantee remain unclaimed. Hosted platform and sanitizer gates are pending. | 28 September 2026 | docs/RELEASE_0_1.md